The Dysthe Project

A plain-language progress report

Work reviewed through 23 September 2026

1. What is this project trying to do?

The project asks how a concentrated group of waves changes as it travels. Does it spread out? Keep its shape? Break into smaller features? What happens when two groups meet? A second question has become increasingly important: can artificial intelligence learn to predict these changes without repeating the full calculation every time?

A model is a simplified set of rules for how a system behaves. A useful picture is a short burst of light rather than an endless, steady beam. The burst contains many fast oscillations. Instead of tracking every one, a wave model can describe the overall shape surrounding them. That shape is called the *envelope*, much as the outline of a musical sound shows when it grows louder and fades.

Dysthe equations describe certain kinds of wave-envelope behavior. The project began with these models and has since branched into plasma, a state of matter containing free electrically charged particles. The connection is the approach to computing and learning physical behavior. The plasma experiment uses its own equations for particle motion and electric fields.

Where the work stands There are now working simulation programs, a collection of checked experiments, two approaches to AI prediction, and tools for running and reviewing large computer jobs. One early AI model matched a short light-wave test closely. The more general plasma model still makes substantial physical errors. The work has therefore produced both useful capabilities and a much clearer picture of what needs improving.

Part of the project	What it studies	What to keep in mind
Light-wave models	How a pulse spreads, concentrates or changes shape.	The main study of light-pulse evolution.
Water-wave model	Traveling pulses and collisions in a simpler setting.	A separate equation and a separate test.
Plasma extension	Electron motion and the electric field it creates.	A new test of AI prediction, using different physics.

How to read this report Pages 2–3 explain the wave simulations. Page 4 introduces the AI work. Page 5 explains the latest plasma results. Page 6 describes the practical outcome and next steps. There are no equations to learn. Every numerical result is from a computer experiment, not a laboratory measurement.

2. First, understand how waves spread

The first calculations used a simplified setting called a *linear* model. Here, small component waves evolve independently and can be added together. This makes it easier to understand what the computer should do before introducing interactions that are harder to predict.

The main tool was *Fourier analysis*: separating a complicated wave into simpler repeating patterns. It is similar to identifying individual notes in a chord. The program advances each component and combines them again to recover the evolving pulse.

What was built A simulator that accepts different starting shapes and tracks their strength, width and movement. The experiments varied the starting pattern, the direction of its oscillations and the model's physical settings. Plots, three-dimensional views and animations made the movement easier to inspect. Separate mathematical tests examined how quickly selected measures of wave strength fall as a pulse spreads.

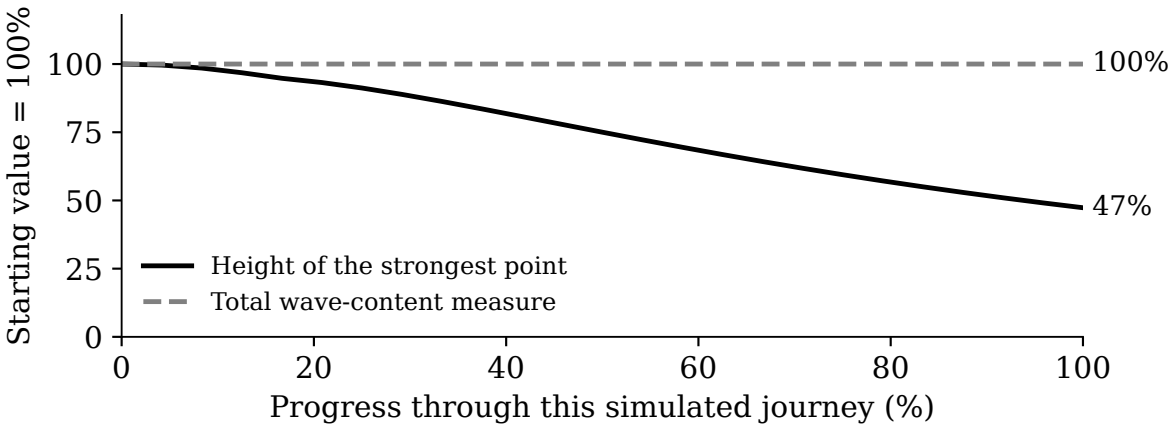


Figure 1: One computed light-wave example. The strongest point drops to about 47% of its starting value while the total wave-content measure stays constant. Both lines are scaled to their own starting values. This is spreading, not evidence that the missing peak strength became heat.

What was learned A lower peak does not mean that the whole wave has disappeared. The same wave content can occupy a wider region, with a smaller highest point. Different starting oscillations can also produce different spreading, even when the initial outlines look identical.

Longer simulations exposed a practical problem: the strongest part of a wave can move far outside the area originally being watched. The project therefore enlarged its search region, increased numerical detail, and compared selected answers with an independent calculation. Four demanding late-time examples passed the chosen checks on their peak values. Additional frequency and specially chosen input tests explored which patterns deserve closer mathematical attention.

What this does not establish These experiments support particular computed examples. They do not prove a rule for every possible wave or for indefinitely long travel. Some earlier results changed when the computational region was enlarged; those needed refinement before interpretation. The plots are evidence to investigate, not a replacement for a mathematical proof. [1]

3. Then, let waves influence their own motion

In a *nonlinear* model, the strength of a wave changes how it evolves. A familiar analogy is traffic: a few cars move almost independently, but a crowded road changes how every driver can move. For waves, these interactions can compete with spreading and sometimes help a pulse keep its shape.

Traveling pulses and collisions The project built a nonlinear simulator and checked it against a simpler case with a known mathematical answer. It then computed 28 traveling-pulse shapes for the water-wave model and tested their motion, small disturbances and collisions.

The collision tests showed that the outgoing pulses did not perfectly recover their original shapes. About 1.8% of the measured wave content appeared outside the two chosen main-pulse regions, beyond what was there initially. That number depends on how the regions are drawn; it is evidence of a small leftover disturbance, not a claim that energy vanished. [2]

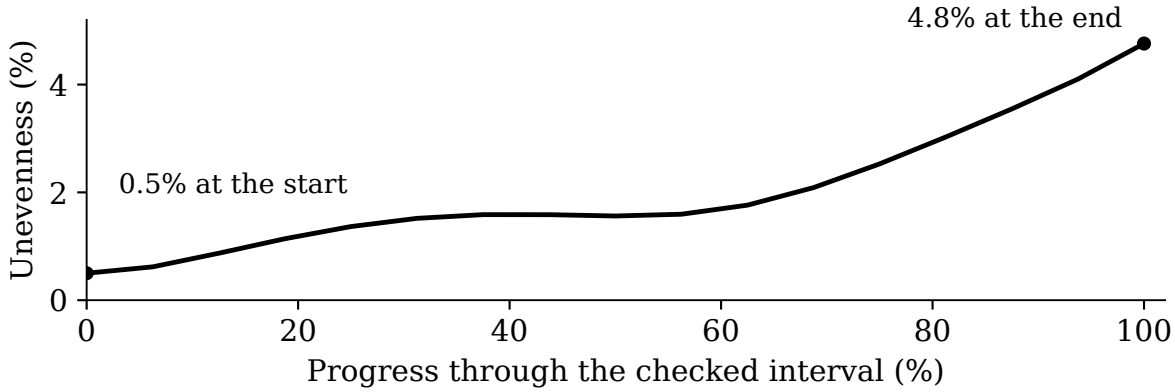


Figure 2: A different experiment, using the light-wave model: a small variation across a stripe-shaped pulse grew from about 0.5% to 4.8%. This measures unevenness in the wave pattern, not a percentage increase in its peak height. Only the interval supported by checks using finer calculations is shown.

Small disturbances can matter The optical experiment shows that a deliberately introduced unevenness can grow. Finer calculations supported the early growth. Later images changed substantially when the numerical detail changed, so they cannot yet establish whether the pulse eventually breaks apart or collapses. Nor do these stripe-shaped examples establish an isolated, stable three-dimensional pulse.

Looking much farther ahead Separate long optical runs showed repeated tightening and loosening that looked like "breathing." These simulations used a repeating box: a wave leaving one side reappears on the opposite side, like a character wrapping around the edge of a video game. Such a box can affect long-term behavior. One coarse preview also stopped after a consistency check failed. The observations are therefore clues to investigate, not proof of a permanently breathing pulse in open space. These long runs were conventional simulations, not successful long-term AI forecasts. [3]

4. Can AI learn a useful shortcut?

A conventional simulator repeatedly applies the model's rules. An AI model instead learns an adjustable approximation. The hope is to obtain useful predictions quickly after training. That speed advantage must still be measured at an acceptable level of accuracy; it has not been established for the current plasma predictor.

The first approach: learn one chosen problem A physics-informed neural network, or *PINN*, was trained to follow the optical equation. The equation acts like a set of rules for marking its answers. The starting pulse was built into the model.

After 20,000 training updates, a run on a research computer had a largest overall wave-pattern mismatch of about **0.67%** over its short test interval. The comparison checked both the strength and oscillation information in the wave. Separate tests supported the numerical reference on that fixed box. This is a useful result for one prescribed starting pulse, not evidence that the network can handle any pulse or predict indefinitely. Changing the starting condition requires retraining this implementation. [3]

The next approach: learn a family of problems A Fourier neural operator, or *FNO*, takes a whole field as input and predicts its next state. A field is simply a map giving a value at every location, like a temperature map. The goal is to learn from many examples and then handle new starting states within the tested range. It reuses Fourier ideas to recognize patterns at different spatial scales. [4]

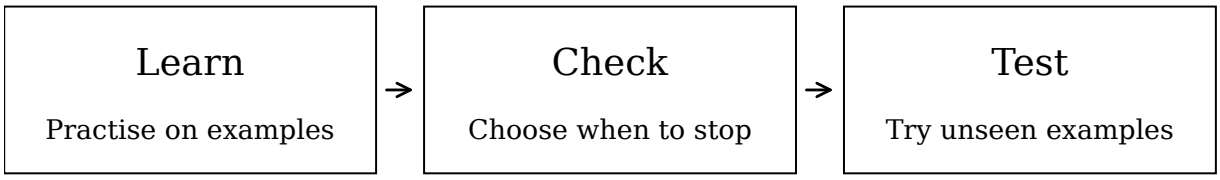


Figure 3: The learning process separates examples used for practice, examples used to choose a model, and examples reserved for its final test. The final test loses that independence if it is repeatedly used to tune the model.

Why more training is not always better The early plasma campaign improved on its practice examples while its longer, self-generated predictions became worse. Feeding each prediction back into the model lets small mistakes accumulate. A good practice score therefore did not guarantee a good forecast.

The follow-up added *early stopping*: stop when the separate checking examples stop improving, and retain the best saved model. It also tested longer prediction sequences during training and added penalties for violating physical rules, such as creating or losing particles or energy. Training logs now show the separate errors, stopping reason and resource use. These changes make mistakes easier to find; they do not automatically make the predictions correct.

5. What did the plasma experiments show?

Plasma contains charged particles whose motion and electric fields affect one another. This project uses a deliberately simplified model: electrons move along one direction while a background of positive charge stays fixed. Rather than tracking every electron, it predicts a map of how many electrons are at each location and moving at each speed.

One test concerns *Landau damping*: a collective electric oscillation weakens as energy transfers into particle motion. Predicting that behavior requires getting the evolving pattern right, not just making a curve go down. The study does not include magnetic fields, collisions or a laboratory plasma experiment. [4]

First campaign: a working model with serious weaknesses Three independently initialized training runs performed better than simply assuming nothing would change. But their particle-motion patterns and energy bookkeeping could be badly wrong, even when the electric-field average looked promising. The best saved models occurred well before the training ended. This motivated the stopping rules and physics checks in the follow-up. [5]

Latest campaign: the tradeoff became clearer The follow-up completed 15 training runs across five setups, then tested the selected setup and a comparison setup in six evaluations. It reserved 44 fresh random cases and four named checks. Total recorded time was about 10.5 hours on one graphics processor.

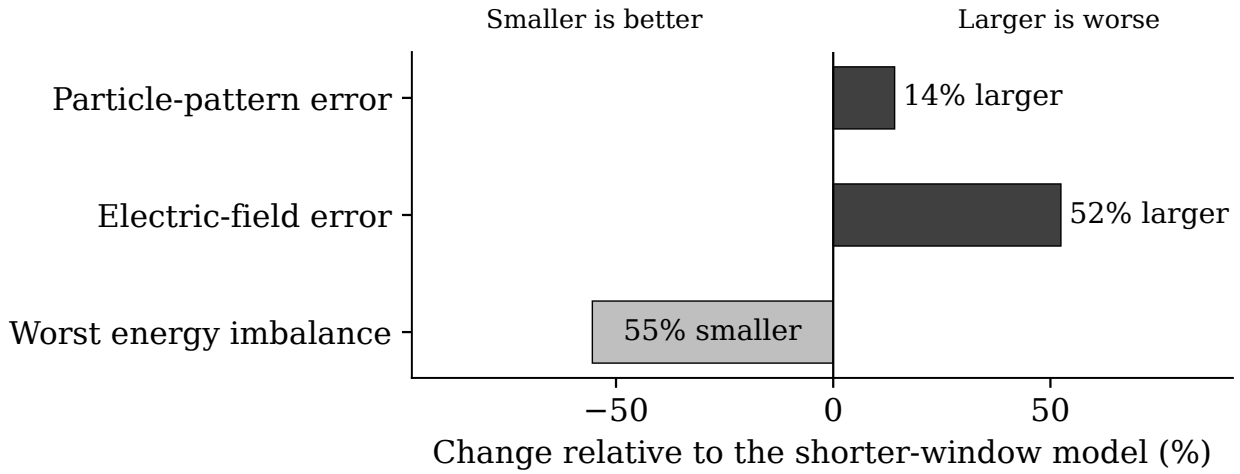


Figure 4: Within the latest campaign, the selected model was trained on longer sequences. It reduced its worst energy imbalance but made larger average particle-pattern and electric-field errors. These are changes in error measures, not percentages of an answer that is "correct."

Its worst energy imbalance fell from about 31% to 14%, still above the study's 5% limit. Every evaluated model failed seven of eight acceptance checks. The selected model's combined checking score had hidden an important tradeoff: improving one physical measure made other predictions worse. A drawing can use the right amount of ink and still show the wrong picture.

The version with an extra penalty for breaking the evolution equation improved that particular checking measure, but reached its time limit while still improving. It was not selected for final testing, so its full potential remains unresolved. The two campaigns also used different test sets and training budgets; their headline percentages are not a controlled before-and-after ranking. [6]

6. What has the project accomplished?

The project has produced a reusable way to investigate wave behavior: write down a particular physical model, compute examples, check the answers, and test whether an AI approximation can reproduce them. The main value is both the software and the evidence about where it can and cannot be trusted.

The practical work behind the results There are now simulation programs for several clearly identified models; checks against known answers and finer calculations; measures of peaks, spreading, shape and physical consistency; and saved results, animations and editable reports. Cluster-job scripts handle larger calculations, training checkpoints allow work to resume, and export tools bring results back for review. The latest reports keep small numerical datasets alongside sharp, zoomable figures so the evidence can be inspected again.

What remains open The work has not demonstrated a universal law for long-term wave decay, a stable isolated three-dimensional pulse, or a broadly reliable plasma AI. The short optical AI result is encouraging within its limited test. The plasma results show why a model must pass several checks at once. None of these studies has yet been validated against laboratory measurements.

The next useful step Improve plasma predictions over short periods before asking for much longer forecasts. Keep pattern accuracy and conservation checks separate when choosing a model. Give the physics-based training a fair comparison, retain enough output to diagnose failures, and use a new final test set once the present results have guided changes. For the wave simulations, longer runs still need checks that distinguish real structure from the repeating box and numerical error.

Useful word	Plain-language meaning
Simulation	A computer calculation of a specified physical model.
Reference	A carefully checked calculation used for comparison, not an infallible answer.
Training step	One adjustment to an AI model; not a moment of physical time.
Conservation	Keeping quantities that the model says should stay constant, such as total energy.
Norm	A numerical ruler for measuring a whole pattern; different rulers emphasize different features.
Validation	Separate checking examples used during model development.

Where the evidence comes from

[1] Project wave-spreading reports, saved histories and spatial-reliability checks, including the September 10 checkpoint.

[2] *Nonlinear Solitary-Wave Report*, September 14, and its saved results. The water-wave model follows Fedele and Dutykh's spatial Dysthe study.

[3] Optical PINN evaluation `evaluation-001`, its separate reference audit, and *Technical Review of the Late-Time Experiment*, September 18.

[4] Background: US Department of Energy, plasma; Li et al., Fourier neural operators; Parker and Dellar, kinetic plasma simulation.

[5] Review of overnight plasma job 77761, September 21.

[6] Technical review of stability study 78140, September 23. The companion source bundle records the numerical inputs and their file hashes. This overview reviews archived work; it does not add new simulation runs.